

Bignonia catalpa syn. *Catalpa bignonioides* (Bignoniaceae)

Catalpa

Description Deciduous tree growing to 65 ft (20 m). Has large oval leaves in whorls of 3, white flowers in conical clusters, and long thin fruits (bean pods).

Habitat & Cultivation Native to the southeastern U.S., this tree is often planted in gardens in southern and western Europe.

Parts Used Bark, fruit.

Constituents The bark contains catalpine, and oxylenzoic and protocathectic acids.

History & Folklore Catalpa bark was formerly used as a substitute for quinine in treating malaria.

Medicinal Actions & Uses The mildly sedative and narcotic bark is used to treat asthma, whooping cough, and other spasmodic coughs in children. The distilled water of the fruit, in combination with herbs commonly used to treat eye problems, such as eyebright (*Euphrasia officinalis*, p. 209) and rue (*Ruta graveolens*, p. 265), makes an effective eyewash for conjunctivitis and other eye infections.

⚠ **Caution** Never use the roots, which are highly poisonous.

Bixa orellana (Bixaceae)

Annatto

Description Evergreen tree growing to 26 ft (8 m). Has large leaves, pink or white flowers, and red fruit capsules containing red seeds.

Habitat & Cultivation Native to tropical forests in the Americas and the West Indies, annatto is widely cultivated in similar climatic zones, notably in India. Seeds are collected as the fruit splits open.

Parts Used Seeds, leaves, root.

Constituents The seed pulp contains carotinoid coloring principles.

History & Folklore In tropical South America, the brilliant red pigment in the seed pulp has traditionally been used in body painting. Annatto dye is also used as a colorant for margarine and cheese.

Medicinal Actions & Uses In the Caribbean, annatto leaves and roots are used to make an astringent infusion that is taken to treat fever, epilepsy, and dysentery. The infusion is also taken as an aphrodisiac. The leaves alone make an infusion that is used as a gargle. The seed pulp reduces the severity of blistering when applied immediately to burns. Taken internally, the seed pulp acts as a general antidote for poisoning.

Borago officinalis (Boraginaceae)

Borage

Description Hairy annual growing to 2 ft (60 cm). Has a pulpy stem, large basal leaves, and attractive blue flowers in summer.

Habitat & Cultivation Borage is a common Mediterranean weed thought to originate from southern Spain and Morocco. Often grown as a garden herb, it is also extensively cultivated for its seed oil.

Parts Used Aerial parts, flowers, seed oil.

Constituents Borage contains mucilage, tannins, and pyrrolizidine alkaloids, which in isolation are toxic to the liver. The seeds contain up to 24% gamma-linolenic acid.



Borage's attractive blue flowers are traditionally used to decorate salads.

History & Folklore The herbalist John Gerard, writing in 1597, extols borage's virtues, "A syrup made of the flowers of borage comforteth the heart, purgeth melancholy, and quieteth the phreneticke or lunaticke person." Gerard also quotes the old saying, "I, Borage, bring always courage."

Medicinal Actions & Uses With its high mucilage content, borage is a demulcent herb and soothes respiratory problems. Its emollient qualities make it helpful for sore and inflamed skin—prepared either as freshly squeezed juice,

in a poultice, or as an infusion. The flowers encourage sweating and the leaves are diuretic. The seed oil is particularly rich in polyunsaturated fats, and is superior in this respect to evening primrose oil (*Oenothera biennis*, p. 241). Borage seed oil is used to treat premenstrual complaints, rheumatic problems, eczema, and other chronic skin conditions.

Cautions Due to the presence of toxic pyrrolizidine alkaloids, there is uncertainty over borage's safety as a medicine. It is subject to legal restrictions in some countries and should not be taken internally. These restrictions and cautions do not apply to borage seed oil.

Boswellia serrata (Burseraceae)

Boswellia, Indian Olibanum, Sallaki (Sanskrit)

Description Deciduous tree, growing to about 49 ft (15 m), with pale papery bark, compound leaves, and clusters of small white flowers. The gum resin exudes from the bark and is transparent gold in color.

Habitat & Cultivation Boswellia grows in dry, hilly regions of central and northern India.

Parts Used Gum resin, bark.

Constituents Boswellia contains triterpene acids (including beta-boswellic acid), essential oil, terpenols, monosaccharides, uronic acids, sterols, and tannins.

History & Folklore Boswellia, a close relative of frankincense (*B. sacra*), has been used for many thousands of years as an astringent and anti-inflammatory.

Medicinal Actions & Uses Boswellia makes an effective mouthwash and gargle, the antiseptic and astringent resin helping to heal and tighten inflamed mucous membranes. Sore throat, laryngitis, canker sores, and gum disease will all benefit. Boswellia is also markedly anti-inflammatory and antiarthritic. This makes it potentially useful in chronic inflammatory diseases such as rheumatoid arthritis, ulcerative colitis, and psoriasis. Other conditions that may benefit include gout, asthma, hay fever, and nettle rash.

Research Research into Boswellia is ongoing and has expanded in recent years. Clinical trials (mostly in India and Germany) indicate that Boswellia counteracts inflammation in conditions such as arthritis and inflammatory bowel disease. The majority of trials noted that symptoms such as pain, stiffness, and poor grip strength all improved. There is increasing evidence that Boswellia has marked pain-relieving activity, and that it promotes stable blood glucose levels in people with type 2 diabetes.